UP504 Applied Spatial Statistics

Motivations for going spatial

(i) Independence assumption not valid:
↳ attributes of observation i may influence attributes of j

(ii) Spatial heterogeneity:
Magnitude and direction of treatment effect may vary across space

(iii) Omitted variable bias:
Some common unobserved or latent variables shared by neighbors

Non-spatial DGP

$$y_i = X_i\beta + e_i \quad \text{with } e_i \sim iid\,(0, \sigma^2)$$

Assumptions

* Observed values at location i independent of values at location j
* Independent residuals, $E[e_i e_j] = E[e_i]E[e_j] = 0$

Spatial DGP

2 neighbours i and j

$$y_i = \alpha_j y_j + X_i\beta + e_i \quad \text{with } e_i \sim iid(0, \sigma^2)$$

$$y_j = \alpha_i y_i + X_j\beta + e_j \quad \text{with } e_j \sim iid(0, \sigma^2)$$

↳ Problem: with more spatial dependencies, little practical usefulness since it results in system with more parameters than observations.

∴ potentially $n^2 - n$ relations to be estimated

Assumptions

* Observations at location i depend on those at location j and vice versa.

* DGP is simultaneous across space

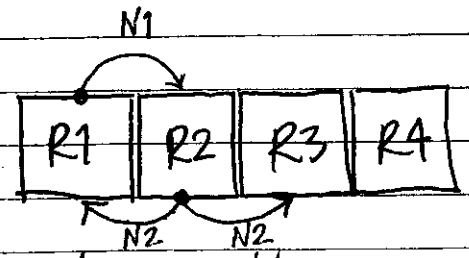
* Overparameterisation problem can be solved by imposing structure in an spatial autoregressive process

spatial correlation coefficient

(2) $$y_i = \rho \underbrace{\sum_{j=1}^{n} W_{ij} y_j}_{} + e_i$$

spatial lag of y_i; weighted average of neighbouring obs.

Example :



↳ first order neighbours and higher order neighbours.
NB: R1 is its own second order neighbour (ie the neighbour of the first-order neighbour)

∴ In matrix notation

$$y = \underset{n\times n}{\rho W}\,\underset{n\times 1}{y} + e \quad \text{with} \quad e \sim N(0, \sigma^2 I_n)$$

In order to construct W, we start with a contiguity matrix C where

$$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Rows ≡ observation i

columns ≡ neighbours j

↳ create W by row-standardisation, ie row sums normalised to one, such that

$$W = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$Wy = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} y_2 \\ (y_1+y_3)/2 \\ (y_2+y_4)/2 \\ y_3 \end{bmatrix}$$

* ρ is a scalar parameter of spatial dependence with

$$\rho \begin{cases} 0 & \text{no spatial autocorrelation} \\ > 0 & \text{pos. SA (clustering)} \\ < 0 & \text{neg. SA (segregation)} \end{cases}$$

$\therefore$ <u>SA DGP:</u>

$$y = \rho W y + \iota\alpha + e$$

$$(I_n - \rho W) y = \iota\alpha + e$$

$$y = (I_n - \rho W)^{-1} \iota\alpha + (I_n - \rho W)^{-1} e$$

$\sum_{k=0}^{\infty} ar^k$ Sum of a geometric series with common ratio $|r|<1$ sum $= \frac{a}{1-r}$

Assuming $|\rho| < 1$, we can express the inverse as

$$(I_n - \rho W)^{-1} = I_n + \rho W + \rho^2 W^2 + \rho^3 W^3$$ which implies

$$y = \iota\alpha + \rho W \iota\alpha + \rho^2 W^2 \iota\alpha + \dots + e + \rho W e + \rho^2 W^2 e + \dots \quad (4)$$

The first part converges to (because α is a scalar and $W_1 = 1$)

$$(1-\rho)^{-1}\iota\alpha \qquad \text{(sum of geometric series)}$$

$$\therefore \quad Y = \frac{1}{(1-\rho)}\iota\alpha + e + \underbrace{\rho We + \rho^2 W^2 e + \ldots}_{\text{violation of Gauss-Markov}}$$

↳ strong violation of Gauss-Markov assumption of independent error terms

$\therefore$ OLS will be <u>biased and inconsistent</u>. Consider following SAR model:

$$Y = \rho WY + X\beta + e$$

$$Y = (I_n - \rho W)^{-1} X\beta + (I - \rho W)^{-1} e$$

$$\text{with } e \sim N(0, \sigma^2 I)$$

↳ Estimate 3 parameters, β, σ^2 and ρ

NB: If $\rho = 0$, SAR reduces to regular OLS model

$$\hat{\beta}_{SAR} = (X'X)^{-1} X'(I - \hat{\rho}W)Y$$

$$\hat{\beta}_{SAR} = (X'X)^{-1} X'Y - \hat{\rho}(X'X)^{-1} X'WY$$

$$= \hat{\beta}_{OLS} - \hat{\rho}(X'X)^{-1} X'WY$$

Global Autocorrelation: Moran's I

* Moran's I coefficient calculates the ratio b/w product of variable and its spatial lag, with product of variable adjusted for spatial weights

$$I = \frac{\dfrac{\sum\sum w_{ij}(y_i-\bar{y})(y_j-\bar{y})}{\sum_i\sum_j w_{ij}}}{\dfrac{\sum(y_i-\bar{y})^2}{n}} = \left[\underset{\text{perfect dispersion}}{-1}, \underset{\text{perfect correlation}}{+1}\right]$$

* Local Moran's I:

Moran's I as the regression coefficient of a variable on its spatial lag.

$$I = \frac{(y_i-\bar{y})\sum w_{ij}(y_j-\bar{y})}{\dfrac{\sum(y_i-\bar{y})^2}{n}}$$

↳ show example of "heat map" of local Moran's I z-scores where values are significantly different from those in neighboring areas

A brief excursion on projections

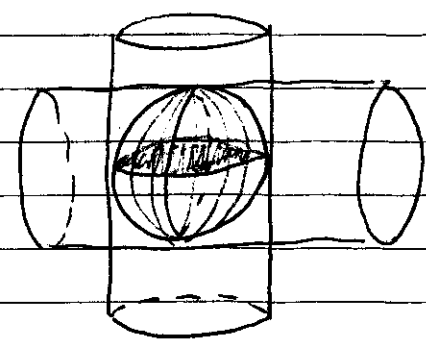
Lat : North-south position ≡ y-axis

Lon : East-west position ≡ x-axis

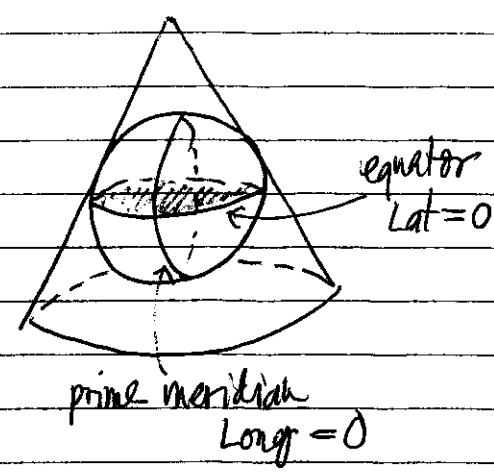
Projection is necessary to represent 3D globe with a 2D map; projections always distort some property of the original globe.

3 main types of projections

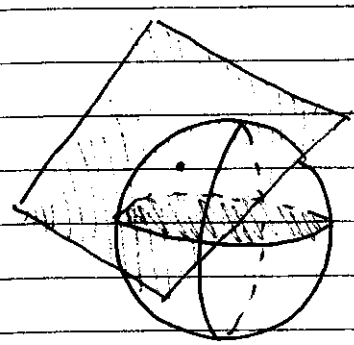
(i) Cylindrical projection:



(ii) Conic projections:



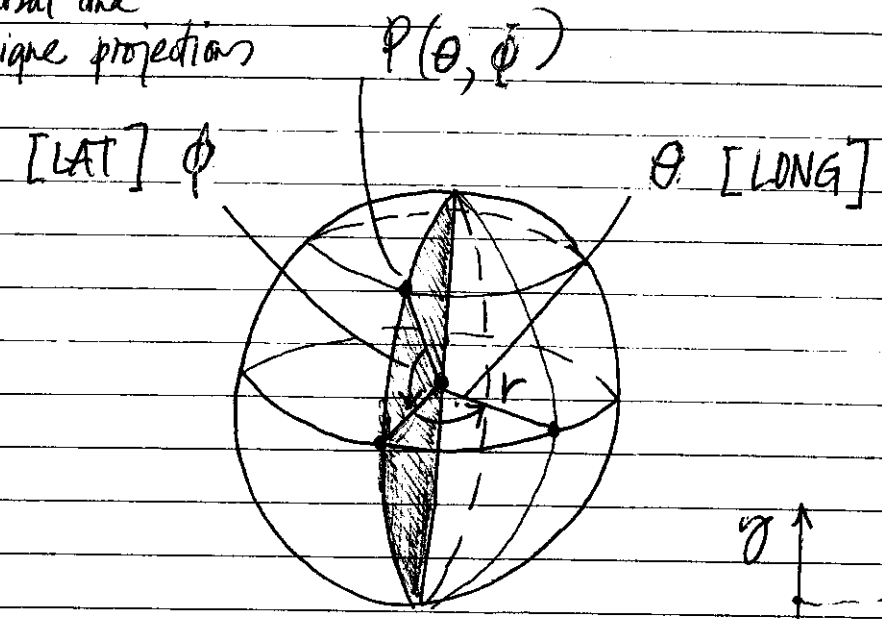
(iii) Planar projections:

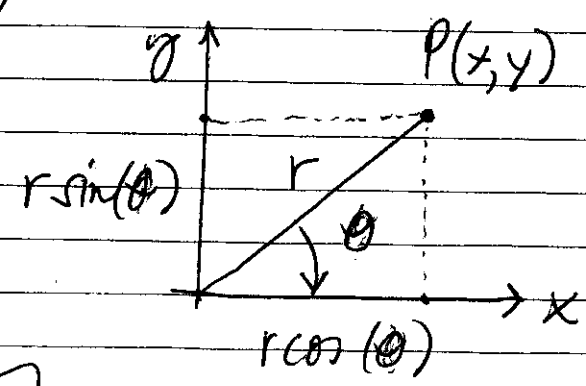


R names of projections and what they preserve:

Preserves	Cylindrical	Conic	Planar
Area	cylequalarea	albers	azequalarea
Angle	mercator* (utm, omerc)	lambert	stereographic
Distance	—	—	azequidistant

* Transversal and oblique projections





Different coordinate systems:

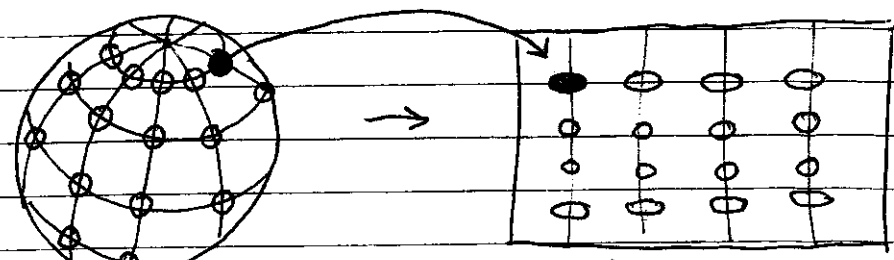
(i) Cartesian coordinates

(ii) Polar coordinates

$\therefore r = \sqrt{y^2 + x^2}$

with $\theta = \tan^{-1}\left(\frac{y}{x}\right)$
$= \arctan\left(\frac{y}{x}\right)$
(depending on quadrant)

Tissot indicatrix ≡ Characterizes distortions due to map projections



Projection
(eg. equirectangular / equidistant cylindrical projection)

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UP 504 <u>Applied Spatial Statistics 2</u>

<u>Kernel density estimation</u>: Nonparametric estimation

* Histogram as most simple NP approach
* Kernel density estimator has two key ingredients:

  (i) <u>bandwidth</u>: width of bin divided by 2, <u>parameter</u> that determines the size of the neighborhood around x_0.

  (ii) <u>kernel function</u>: a weight-assigning or smoothing

∴ general kernel probability density estimator function:

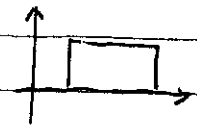
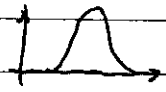
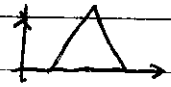
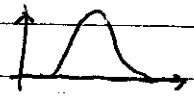
$$pdf(x_0) = \frac{1}{Nh} \sum_{i=1}^{N} K\left(\frac{x_i - x_0}{h}\right)$$

↳ histogram estimator as rectangular kernel density estimator with

$$K(x) = \begin{cases} 1/2 & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

with bandwidth $h = 1/2$ (→ Bertillon's data of height by 1 inch buckets)

Alternative Kernels:

- rectangular (assigns identical weights across bin) 
- Epanechnikov
- Biweight 
- Triangular 
- Gaussian (normal) 

eg. Gaussian Kernel density estimator function:

$$\hat{f}(x) = \frac{1}{Nh} \sum_{i=1}^{N} \frac{1}{\sqrt{2\pi}} \exp\left(-0.5 \cdot \left[\frac{x_i - x_0}{h}\right]^2\right)$$

<u>NB</u>: Bins can be overlapping (histogram has non-overlapping bins) ∴ most kernels let the weights decline the further the observation is from the center of the bin, ie. x_0.

↳ <u>Choice of bandwidth</u> more important in practice than choice of kernel; essentially involves a trade-off b/w:

* bias (misreporting shape)
* variance (lack of precision)

Estimating models with SAR DGP

Recall: OLS is inconsistent and biased

$$\hat{\beta}_{SAR} = \hat{\beta}_{OLS} - \hat{\rho}(X'X)^{-1}X'Wy$$

∴ Use alternative estimators: MLE or IV 2SLS

↳ 2SLS has similar properties to MLE (consistency and asymptotic normality) but is a lot easier to implement numerically.

Eg. Address endogeneity due to spatial variable linkage and error term by using lagged variables as instruments

Let $X = [Wy \;\; X_1]$ and $Z = [L(y) \;\; X_1]$

$$\therefore \hat{\beta}_{SAR_IV} = [X'P_Z X]^{-1} X'P_Z y$$

with $P_Z \equiv Z[Z'Z]^{-1}Z'$

Different spatial models: (relative to OLS: $y = X\beta + e$)

(i) SAR: $y = \rho Wy + X\beta + e$

(ii) SEM: $y = X\beta + (I - \lambda W)^{-1} e$ → with $u = \rho Wu + e$ spatial error model

(iii) SDM: $y = \rho Wy + X\beta + WX\theta + e$ spatial Durbin model

* Extensions to basic spatial models:

(i) Spatial autoregressive moving average model: SARMA (like SAL plus different error term series)

(ii) Spatial autocorrelation model (SAC) (spatial dependence in y and e, with different weight matrices W_1 and W_2)

(iii) Spatial Durbin error model (SDEM)

Geographically weighted regression (GWR)

Linear model as $y_i = X\beta_i + e$ with locally-varying estimates for β

∴ Parameter estimates use non-parametric local weighting scheme w_{ij}:

$$\beta_{GWR} = (X'W_iX)^{-1}X'W_iy,$$

where weight w_{ij} for observation j is calculated with a decay function (and a Gaussian kernel)

$$w_{ij} = e^{\left(-d_{ij}/h\right)^2}$$

where $d_{ij} \equiv$ Euclidean dist, $h \equiv$ user selected bandwidth